Supplementary Material for the Paper "Learning to Compute Products of Polynomials with Transformers" *

Yuxuan Song[1,2], Bin Wang[1,2], and Changbo Chen[1,2]

[1] Chongqing Institute of Green and Intelligent Technology, Chinese Academy of Sciences
Chongqing, China
[2] Chongqing School, University of Chinese Academy of Sciences
Chongqing, China

In this supplementary material, we provide some additional experimental results.

1 The impact of tokenization scheme

Table 1 evaluates the performance of Strategy 1 (Direct Strategy) trained with four different tokenization schemes based on a comprehensive set of evaluation metrics. We have the following observations: (*i*) Different tokenization schemes have a big impact on the performance of Transformer models on the dataset v1d3C1 and v1d7C1 and the overall best one is ImDs; (*ii*) Even with the best tokenization scheme, Direct Strategy still fails to predict the exact product on the most challenging dataset v1d7C1; (*iii*) No matter which tokenization scheme is used, the Transformer models can predict the legal expression, the degree, the number of terms in nearly perfect accuracy. The support can also be predicted in a high accuracy.

Table 1: Evaluation of Strategy 1 with four different tokenization schemes.

Dataset	Tokenization	TA	SA	LA	DA	NA	SPA	SSA	C1A	C2A	MA	CA
v1d1C1	ImUs	99.98	99.95	**100**	**100**	**100**	**100**	**100**	99.98	99.95	99.95	99.95
	ImDs	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
	ExUs	99.99	99.97	**100**	**100**	**100**	**100**	**100**	99.97	99.97	99.97	99.97
	ExDs	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
v1d3C1	ImUs	92.22	23.84	**100**	99.97	**100**	98.58	93.93	49.83	24.89	23.84	23.84
	ImDs	**94.49**	**70.93**	**100**	**100**	**100**	**99.40**	**96.17**	**82.23**	**71.86**	**70.93**	**70.93**
	ExUs	92.17	4.24	**100**	99.94	**100**	97.62	87.53	14.90	4.65	4.24	4.24
	ExDs	85.46	29.54	**100**	99.98	**100**	98.60	92.93	52.78	30.75	29.54	29.54
v1d7C1	ImUs	64.92	0	99.94	99.61	99.94	90.68	45.53	0	0	0	0
	ImDs	56.70	0	**100**	99.99	**100**	**94.36**	**85.88**	**9.48**	**0.05**	0	0
	ExUs	**69.83**	0	99.97	99.82	99.97	89.34	0.16	0	0	0	0
	ExDs	6.94	0	99.99	99.89	99.99	90.20	0.20	0	0	0	0
v2t2E1C1	ImUs	98.16	94.66	**100**	99.94	**100**	99.71	99.66	99.44	99.42	99.42	98.57
	ImDs	98.05	96.22	99.99	99.95	99.99	99.84	**99.82**	**99.74**	**99.73**	**99.73**	**98.83**
	ExUs	**98.79**	**96.47**	99.99	99.98	99.99	**99.86**	99.79	98.91	98.84	98.84	98.11
	ExDs	97.33	93.88	**100**	99.40	**100**	98.31	98.27	97.03	96.86	96.86	96.05
v3t3E1C1	ImUs	96.03	65.54	**100**	99.91	**100**	98.53	98.50	98.27	98.27	98.27	97.85
	ImDs	95.54	76.55	**100**	**99.96**	**100**	**99.61**	**99.56**	**99.49**	**99.48**	**99.48**	**99.07**
	ExUs	**97.56**	72.88	**100**	99.93	**100**	99.32	94.94	94.42	94.41	94.40	94.00
	ExDs	95.85	**77.07**	**100**	99.94	**100**	99.42	97.33	97.25	97.24	97.23	96.83

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Table 2 further evaluates the performance of Strategy 2 (Collect-Combine) trained with four different tokenization schemes on the dataset v1d3C1 and v1d7C1. We have similar observations as for Table 1 except that overall best tokenization scheme is ExDs. To make the final choice between ImDs and ExDs, Table 3 evaluates the performance of Strategy 3 (Collect-Accumulate) trained with these two schemes on the dataset v1d3C1 and v1d7C1. The experimental results are clearly in favor of ExDs. Note that in Table 3, we observe something unusual for the tokenization scheme ImDs. On both datasets, nearly perfect MA and CA are achieved but the token accuracy TA is quite low. Further investigation shows that under ImDs, the label p (like $2*x^2-2*y^2$) is frequently predicted as $+p$ (like $+2*x^2-2*y^2$). Such prediction is treated to be false when calculating the token and the sequence accuracy as its length is different from that of the label. On the other hand, if the tokenization scheme is ExDs, for the polynomial $p=2*x^2-2*y^2$, it will be always encoded as $+2*x^2-2*y^2$ in the label, which helps reducing the token prediction error rate.

Table 2: Evaluation of Strategy 2 with four different tokenization schemes.

Dataset	Tokenization	TA	SA	LA	DA	NA	SPA	SSA	C1A	C2A	MA	CA
v1d3C1	ImUs	90.40	20.41	99.99	99.96	98.33	98.28	52.33	24.93	20.56	20.41	20.41
	ImDs	97.96	93.77	**100**	99.98	99.54	99.52	98.66	94.65	93.79	93.78	93.78
	ExUs	91.38	6.60	99.99	99.97	97.97	97.91	37.08	11.38	6.86	6.60	6.60
	ExDs	97.70	90.20	**100**	**100**	**99.55**	**99.54**	98.65	92.60	90.26	90.20	90.20
v1d7C1	ImUs	61.99	0	**100**	99.94	**90.37**	**90.03**	1.69	0	0	0	0
	ImDs	4.16	0	**100**	99.80	82.65	81.73	0.33	0	0	0	0
	ExUs	**66.93**	0	99.99	99.10	99.99	87.67	0.09	0	0	0	0
	ExDs	5.15	0	**100**	90.00	74.08	72.62	0.30	0	0	0	0

Table 3: Evaluation of Strategy 3 with two different tokenization schemes.

Dataset	Tokenization	TA	SA	LA	DA	NA	SPA	SSA	C1A	C2A	MA	CA
v1d3C1	ImDs	44.91	44.88	**100**	99.98	99.90	99.89	99.82	99.74	99.72	99.72	99.72
	ExDs	**95.09**	**95.06**	**100**	**100**	**99.98**	**99.98**	**99.94**	**99.91**	**99.91**	**99.91**	**99.91**
v1d7C1	ImDs	37.74	37.71	**100**	**99.97**	99.88	99.88	99.67	99.54	99.53	99.53	99.53
	ExDs	**90.24**	**90.14**	**100**	99.96	**99.94**	**99.93**	**99.70**	**99.58**	**99.57**	**99.57**	**99.57**

2 The challenge of handling longer polynomials

Our method for handling long polynomials applies the divide-and-conquer approach and combines three different types of models, namely the collected product model, the accumulation model, and the collected sum model. Previous experimental results have demonstrated the performance of the first two types of models. Table 4 shows that the collected sum model also achieves nearly perfect accuracy on all evaluation metrics. In addition, when combing this model with the accumulation model, polynomial addition can be well learned. Finally, Table 5 reports how a combination of these models performs when dealing with long polynomials multiplication. The results show that although the collected product and collected sum can be learned in a high accuracy, it is still challenging to learn the final product of long polynomial multiplication due to the occurrence of long integer addition.

Table 4: Evaluation of learning polynomial addition.

Dataset	Tokenization	Step	TA	SA	LA	DA	NA	SPA	SSA	C1A	C2A	MA	CA
v1d3C2	ExDs	Collected sum	99.84	99.77	100	100	100	99.99	99.92	99.79	99.77	99.77	99.77
		Overall	97.87	97.78	100	100	100	99.99	99.92	99.79	99.77	99.77	99.77
v1d7C2	ExDs	Collected sum	99.89	99.80	99.99	99.99	99.99	99.99	99.93	99.84	99.80	99.80	99.80
		Overall	95.95	95.89	99.99	99.99	99.99	99.99	99.93	99.84	99.80	99.80	99.80

Table 5: Evaluation of learning long polynomials multiplication.

Dataset	Tokenization	Step	TA	SA	LA	DA	NA	SPA	SSA	C1A	C2A	MA	CA
v1d15C1	ExDs	Collected product	94.77	91.22	99.97	99.78	99.97	99.07	92.92	91.39	91.25	91.22	91.22
		Collected sum	99.79	99.72	100	100	100	99.94	99.86	99.76	99.73	99.72	99.72
		Overall	68.64	45.87	100	99.84	100	86.80	72.45	57.15	48.98	47.24	47.24